

UNIT I INTRODUCTION TO R PROGRAMMING

1.STUDENT ACADEMIC PERFORMANCE SYSTEM

A college maintains records of students including student ID, name, marks in Mathematics, Physics, and Chemistry, and overall pass/fail status. The management wants to analyze student performance using R programming. They need to calculate subject-wise averages, total marks, identify students who passed all subjects, and store categorical information properly. The data contains both numeric and non-numeric values and requires appropriate structuring using R data structures.

Questions:

1. Identify suitable data types for each variable and explain why they are appropriate.
2. Create numeric, character, and logical vectors to represent marks, names, and pass status.
3. Combine the vectors into a data frame and explain its structure.
4. Represent subject marks using a matrix and perform row-wise and column-wise calculations.
5. Convert pass/fail status into a factor and explain the importance of factors in R.
6. Explain atomic vectors and character vectors with reference to this case study.
7. Describe how R sessions manage variables and how objects can be removed from memory.

2. HOSPITAL PATIENT MONITORING SYSTEM

A hospital records patient information including patient ID, age, gender, blood pressure readings, and diabetic status (Yes/No). The hospital wants to identify high-risk patients, calculate average blood pressure, and analyze health trends over multiple days. The dataset contains mixed data types and multi-dimensional data collected over time.

Questions:

1. Specify appropriate data types for each variable and justify your answer.
2. Store the data using a data frame and convert categorical variables into factors.
3. Use logical vector operations to identify patients with blood pressure above 140 and diabetic status as “Yes”.
4. Create a list to store mixed data types and explain how it differs from an atomic vector.
5. Represent multi-day blood pressure readings using an array and perform calculations such as mean and maximum values.
6. Explain the role of basic mathematical operations in analyzing this dataset.

3. E-COMMERCE SALES ANALYSIS

An online store records daily sales of four products for seven days. The company wants to calculate total sales, product-wise revenue, and identify days where sales exceeded the weekly average. The dataset consists entirely of numeric values but also includes product names and categories.

Questions:

1. Create a matrix to represent daily sales and perform row and column operations.
2. Multiply the sales matrix by a price vector to calculate total revenue.
3. Store product names as a character vector and explain its properties.
4. Use logical vectors to identify days where sales are greater than the average sales.
5. Explain the difference between vectors, matrices, and arrays using this case study.
6. Define an S3 class called “Product” and explain its structure.

4. WEATHER MONITORING SYSTEM

A meteorological department collects monthly data including temperature, humidity, rainfall, and wind speed for one year. The department wants to analyze seasonal

averages, maximum rainfall, and trends across multiple weather parameters. The dataset involves multi-dimensional measurements recorded over time.

Questions:

1. Create appropriate vectors for each weather parameter and combine them into a data frame.
2. Convert month names into factors and explain why factor is suitable.
3. Create a three-dimensional array to represent multiple weather parameters across months.
4. Perform calculations to determine mean temperature and maximum rainfall.
5. Write a function in R to calculate seasonal average temperature.
6. Explain how arrays differ from matrices.

BANKING LOAN RISK CLASSIFICATION SYSTEM

A bank collects information from loan applicants including age, income, credit score, employment status, and loan approval status (Yes/No). The bank wants to analyze applicant eligibility, classify risk levels, and structure the data efficiently using object-oriented concepts in R.

Questions:

1. Identify appropriate data types for each variable and justify your answer.
2. Store the data in a data frame and convert categorical variables into factors.
3. Use logical operations to identify applicants with income greater than 50,000 and credit score above 700.
4. Create a matrix of financial indicators and perform matrix operations.
5. Define an S4 class called “Loan Applicant” with appropriate slots.
6. Explain the difference between S3 class, S4 class, and Reference class in R.

